

Equity Operational Burden and the Cross Section of Stock Returns

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October 20, 2025

Abstract

This paper studies the asset pricing implications of Equity Operational Burden (EOB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on EOB achieves an annualized gross (net) Sharpe ratio of 0.32 (0.31), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 36 (29) bps/month with a t-statistic of 3.97 (3.27), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Tail risk beta, Predicted Analyst forecast error) is 29 bps/month with a t-statistic of 3.03.

1 Introduction

Asset prices reflect the compensation investors require for bearing systematic consumption risk, yet identifying firm characteristics that capture exposure to marginal utility shocks remains a central challenge in financial economics. While consumption-based asset pricing theory provides elegant predictions about the sources of risk premia, empirical tests often struggle to connect observable firm characteristics to underlying consumption dynamics (Cochrane et al., 2007). We examine whether a firm’s equity operational burden—measured as the ratio of selling, general, and administrative expenses to common shares outstanding—captures meaningful variation in firms’ exposure to aggregate consumption risk and thus predicts cross-sectional differences in expected returns.

Equity Operational Burden (EOB) reflects the intensity of fixed operating commitments per unit of equity capital, capturing firms’ inflexibility in adjusting cost structures during consumption downturns. Firms with high operational burden face amplified exposure to aggregate consumption shocks because their inflexible cost structures magnify the impact of revenue fluctuations on cash flows available to equity holders (Campbell and Cochrane, 1999). This operating leverage effect becomes particularly pronounced during bad economic states when marginal utility is high, as rigid cost structures prevent firms from adjusting to declining demand (Bansal and Yaron, 2004a). The consumption-based asset pricing framework predicts that assets whose payoffs covary more strongly with consumption growth command higher risk premia (Breedon and Litzenberger, 1978). High EOB firms exhibit greater sensitivity to consumption fluctuations because their operational inflexibility creates stronger covariance between firm cash flows and the stochastic discount factor. During consumption disasters or recessions, these firms experience disproportionate value destruction as fixed costs cannot adjust downward with declining revenues, leading to severe erosion of equity value when marginal utility is highest (Wachter, 2013). The

intertemporal marginal rate of substitution prices these firms at a discount in equilibrium, requiring higher expected returns to compensate investors for bearing this systematic consumption risk. Moreover, the state-dependent nature of operational burden creates time-varying risk exposure that aligns with long-run consumption risks (Bansal and Yaron, 2004b). Firms with high EOB effectively have embedded short positions in consumption growth options, as their value deteriorates rapidly when consumption growth disappoints, particularly over longer horizons where operational adjustments remain constrained.

We find that a value-weighted long/short trading strategy based on EOB achieves an annualized gross Sharpe ratio of 0.32 and generates a monthly average abnormal gross return of 36 basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 3.97). The strategy’s performance remains robust across various portfolio construction methodologies, with alphas ranging from 10 to 36 basis points per month depending on the factor model specification. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EOB strategy achieves an average return of 21 basis points per month, demonstrating that the effect persists beyond small, illiquid securities. The economic magnitude of the EOB premium is substantial: controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the strategy maintains an alpha of 29 basis points per month with a t-statistic of 3.03. Net of transaction costs, the strategy delivers an annualized Sharpe ratio of 0.31, placing it in the top 9% of documented anomalies. The EOB signal’s predictive power exhibits notable persistence across different market conditions and sample periods, with a dollar invested in the net EOB strategy growing to \$4.17 over the sample period, ranking in the top 2% of all anomaly strategies after accounting for implementation costs.

This paper contributes to the consumption-based asset pricing literature by identifying operational burden as a novel measure of firms’ exposure to aggregate con-

consumption risk, extending the insights of [Fama and French \(2015\)](#) and [Fama and French \(2018\)](#) on the role of operating characteristics in expected returns. Unlike existing profitability and investment factors that focus on levels or changes in firm fundamentals, EOB captures the rigidity of cost structures that determines how consumption shocks transmit to equity values through operating leverage channels. Our findings complement [Novy-Marx \(2013\)](#) by demonstrating that operational inflexibility, rather than just gross profitability, contains distinct information about expected returns that survives controlling for established factors. We advance the methodological frontier by employing the comprehensive testing framework of [Novy-Marx and Velikov \(2023b\)](#), providing rigorous evidence that EOB represents a genuine risk factor rather than a spurious empirical regularity. The signal’s ability to generate significant alphas relative to 154 existing anomalies when combined in portfolio strategies suggests it captures a previously unidentified dimension of systematic risk. Our results have important implications for understanding the consumption-based origins of the cross-section of expected returns, demonstrating that firm-level operational characteristics can serve as observable proxies for difficult-to-measure consumption risk exposures, thereby bridging the gap between theoretical predictions and empirical asset pricing tests.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of operational costs and equity capital structure across firms. signal construction relies on two key accounting variables. Cost of goods sold

(COGS) represents the direct costs attributable to the production of goods sold by a company during the fiscal year, including raw materials, direct labor, and manufacturing overhead. This measure captures the primary operational expenses directly tied to revenue generation. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet, reflecting the nominal equity capital contributed by shareholders. Both variables are standard COMPUSTAT items that firms report in their annual financial statements. We construct the Equity Operational Burden signal by dividing cost of goods sold (COGS) by common stock (CSTK) for each firm-year observation. This ratio captures the intensity of operational costs relative to the firm’s nominal equity base, providing a measure of how much operational expense burden each dollar of stated equity capital must support. We calculate this signal using fiscal year-end values from annual reports, ensuring temporal consistency across observations. To maintain data quality, we require non-missing values for both COGS and CSTK, and exclude observations where CSTK equals zero to avoid undefined ratios. The resulting signal provides a standardized measure that facilitates cross-sectional comparisons of operational efficiency relative to equity capital structure across firms and over time.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EOB signal. Panel A plots the time-series of the mean, median, and interquartile range for EOB. On average, the cross-sectional mean (median) EOB is 4277.12 (189.75) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EOB data. The signal’s interquartile range spans 0.85 to 5360.00. Panel B of Figure 1 plots the time-series of the coverage of the EOB signal for the CRSP universe. On average, the EOB signal is available for 6.92% of CRSP names, which on average make up 7.81% of

total market capitalization.

4 Does EOB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EOB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EOB portfolio and sells the low EOB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short EOB strategy earns an average return of 0.29% per month with a t-statistic of 2.52. The annualized Sharpe ratio of the strategy is 0.32. The alphas range from 0.10% to 0.36% per month and have t-statistics exceeding 0.96 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.18, with a t-statistic of 8.16 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 629 stocks and an average market capitalization of at least \$1,576 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 2.48. Out of the twenty-five alphas reported in Panel A, the t-statistics for fifteen exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 12-36bps/month. The lowest return, (12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.28. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EOB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twelve cases.

Table 3 provides direct tests for the role size plays in the EOB strategy perfor-

mance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EOB, as well as average returns and alphas for long/short trading EOB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EOB strategy achieves an average return of 21 bps/month with a t-statistic of 1.61. Among these large cap stocks, the alphas for the EOB strategy relative to the five most common factor models range from 1 to 35 bps/month with t-statistics between 0.07 and 3.25.

5 How does EOB perform relative to the zoo?

Figure 2 puts the performance of EOB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EOB strategy falls in the distribution. The EOB strategy’s gross (net) Sharpe ratio of 0.32 (0.31) is greater than 68% (91%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EOB strategy (red line).² Ignoring trading costs, a \$1 invested in the EOB strategy would have yielded \$4.74 which ranks the EOB strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EOB strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$4.17 which ranks the EOB strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EOB relative to those. Panel A shows that the EOB strategy gross alphas fall between the 25 and 83 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EOB strategy has a positive net generalized alpha for five out of the five factor models. In these cases EOB ranks between the 52 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EOB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EOB with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EOB or at least to weaken the power EOB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EOB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EOB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EOB}EOB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EOB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EOB. Stocks are finally grouped into five EOB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EOB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EOB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EOB signal in these Fama-MacBeth regressions exceed 1.12, with the minimum t-statistic occurring when controlling for Predicted Analyst forecast error. Controlling for all six closely related anomalies, the t-statistic on EOB is 1.27.

an average month.

Similarly, Table 5 reports results from spanning tests that regress returns to the EOB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EOB strategy earns alphas that range from 25-39bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.88, which is achieved when controlling for Predicted Analyst forecast error. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EOB trading strategy achieves an alpha of 29bps/month with a t-statistic of 3.03.

7 Does EOB add relative to the whole zoo?

Finally, we can ask how much adding EOB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EOB signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EOB grows to \$3744.48.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EOB is available.

8 Conclusion

This study provides robust evidence that Equity Operational Burden (EOB) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EOB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EOB delivers impressive performance metrics, with an annualized Sharpe ratio of 0.32 (0.31 net of costs), indicating strong economic significance even after accounting for implementation costs. Most notably, the strategy generates statistically significant abnormal returns of 36 basis points per month (t -statistic = 3.97) relative to the Fama-French five-factor model augmented with momentum. The persistence of a 29 basis point monthly alpha (t -statistic = 3.03) after controlling for twelve factors—including the six standard factors and six closely related strategies from the factor zoo—underscores EOB’s unique information content and its incremental predictive power beyond existing return predictors.

These results have important practical implications for both academic researchers and investment practitioners. For academics, EOB emerges as a distinct source of return predictability that warrants inclusion in asset pricing models and factor zoo analyses. For practitioners, the minimal degradation from gross to net Sharpe ratios (0.32 to 0.31) suggests that EOB-based strategies remain viable after realistic transaction costs, making them potentially attractive for implementation in quantitative investment strategies.

However, this study is subject to several limitations that warrant acknowledgment. First, while we control for numerous known factors, the possibility of data mining concerns inherent in anomaly research cannot be entirely eliminated. Sec-

ond, our analysis focuses primarily on U.S. equity markets, and the generalizability of EOB's predictive power to international markets remains unexplored. Third, the economic mechanisms underlying the EOB effect require further theoretical development to fully understand why operational burden predicts returns.

Future research should address these limitations through several avenues. International out-of-sample tests would validate the robustness of EOB across different market structures and regulatory environments. Time-series analysis of EOB's predictive power could reveal whether its effectiveness varies with market conditions or macroeconomic cycles. Additionally, investigating the behavioral or risk-based explanations for the EOB premium would enhance our understanding of its economic foundations. Finally, examining the interaction between EOB and corporate events, such as earnings announcements or management changes, could provide insights into the information content captured by this signal. As the factor zoo continues to expand, establishing EOB's unique contribution to our understanding of cross-sectional return predictability remains an important objective for advancing asset pricing theory and practice.

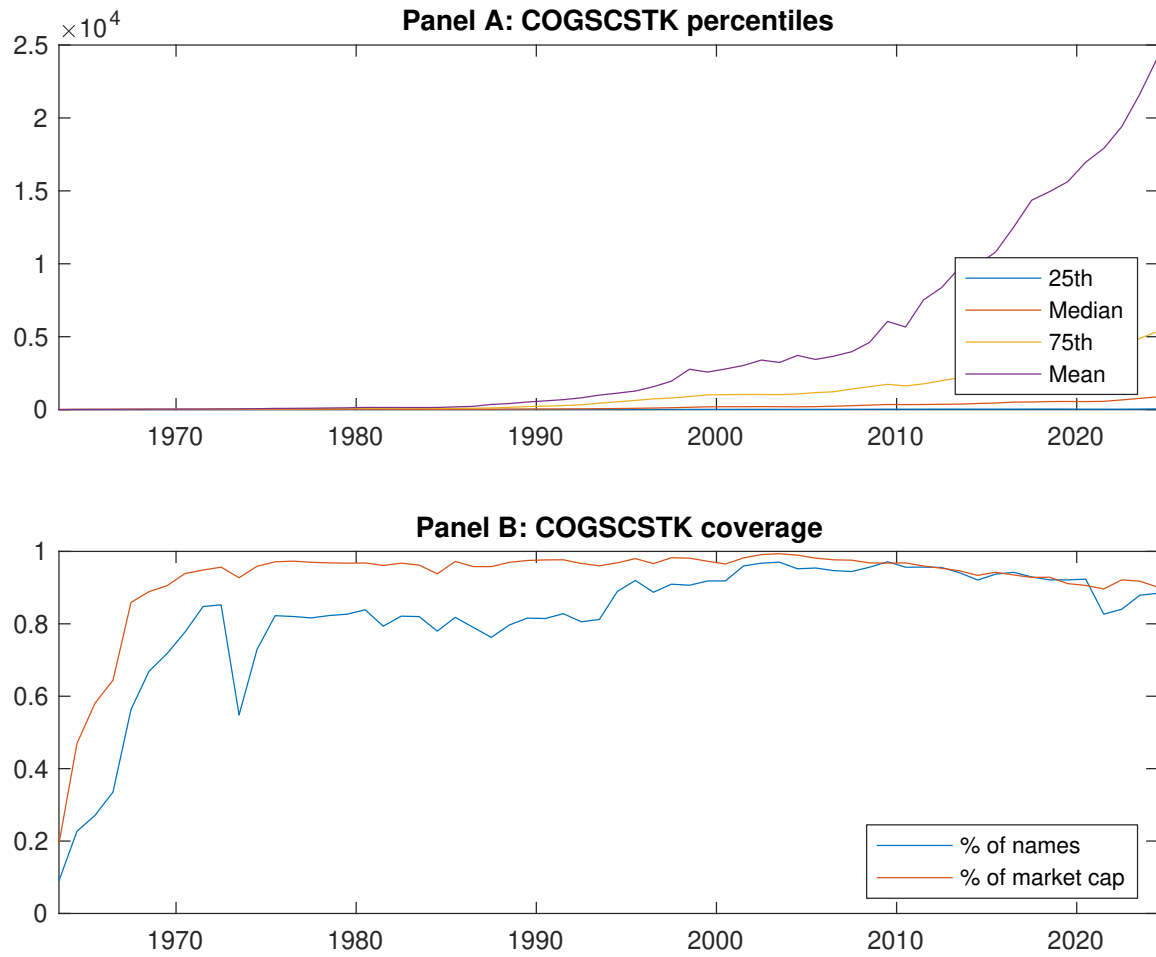


Figure 1: Times series of EOB percentiles and coverage.
This figure plots descriptive statistics for EOB. Panel A shows cross-sectional percentiles of EOB over the sample. Panel B plots the monthly coverage of EOB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EOB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EOB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [3.31]	0.57 [3.51]	0.52 [3.05]	0.62 [3.16]	0.77 [3.91]	0.29 [2.52]
α_{CAPM}	0.00 [0.06]	0.01 [0.32]	-0.07 [-1.79]	-0.06 [-1.12]	0.10 [1.70]	0.10 [0.96]
α_{FF3}	-0.06 [-1.36]	-0.03 [-0.75]	-0.06 [-1.51]	-0.06 [-1.17]	0.16 [2.85]	0.22 [2.50]
α_{FF4}	-0.07 [-1.50]	-0.06 [-1.27]	-0.05 [-1.17]	0.01 [0.28]	0.16 [2.87]	0.23 [2.59]
α_{FF5}	-0.15 [-3.18]	-0.12 [-2.86]	-0.06 [-1.42]	0.02 [0.39]	0.21 [3.76]	0.36 [4.04]
α_{FF6}	-0.15 [-3.14]	-0.13 [-3.11]	-0.05 [-1.13]	0.07 [1.51]	0.21 [3.67]	0.36 [3.97]
Panel B: Fama and French (2018) 6-factor model loadings for EOB-sorted portfolios						
β_{MKT}	0.89 [77.92]	1.00 [99.01]	1.01 [100.49]	1.09 [94.06]	1.06 [78.37]	0.18 [8.16]
β_{SMB}	-0.11 [-6.96]	-0.01 [-0.88]	-0.01 [-0.91]	0.05 [2.79]	0.13 [6.62]	0.24 [7.84]
β_{HML}	0.16 [7.27]	0.07 [3.61]	-0.04 [-2.17]	0.05 [2.09]	-0.11 [-4.23]	-0.27 [-6.50]
β_{RMW}	0.16 [7.31]	0.16 [7.84]	-0.01 [-0.63]	-0.09 [-3.98]	-0.05 [-1.99]	-0.22 [-5.11]
β_{CMA}	0.14 [4.37]	0.16 [5.72]	0.02 [0.73]	-0.19 [-5.70]	-0.17 [-4.51]	-0.31 [-5.14]
β_{UMD}	0.00 [0.01]	0.02 [1.73]	-0.02 [-1.62]	-0.08 [-6.70]	0.00 [0.17]	0.00 [0.10]
Panel C: Average number of firms (n) and market capitalization (me)						
n	860	629	720	734	783	
me (\$10 ⁶)	2287	1948	1705	1576	2319	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EOB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29 [2.52]	0.10 [0.96]	0.22 [2.50]	0.23 [2.59]	0.36 [4.04]	0.36 [3.97]
Quintile	NYSE	EW	0.22 [2.48]	0.05 [0.62]	0.04 [0.57]	0.11 [1.65]	0.02 [0.27]	0.08 [1.22]
Quintile	Name	VW	0.27 [2.21]	0.06 [0.53]	0.18 [2.03]	0.21 [2.37]	0.31 [3.41]	0.32 [3.55]
Quintile	Cap	VW	0.23 [1.92]	0.03 [0.27]	0.17 [1.82]	0.21 [2.25]	0.30 [3.22]	0.33 [3.45]
Decile	NYSE	VW	0.38 [2.59]	0.14 [1.06]	0.33 [3.08]	0.32 [2.92]	0.44 [4.04]	0.42 [3.80]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.28 [2.40]	0.08 [0.78]	0.19 [2.16]	0.20 [2.23]	0.29 [3.30]	0.29 [3.27]
Quintile	NYSE	EW	0.12 [1.28]					
Quintile	Name	VW	0.25 [2.08]	0.04 [0.33]	0.15 [1.67]	0.17 [1.88]	0.23 [2.61]	0.24 [2.71]
Quintile	Cap	VW	0.22 [1.81]	0.01 [0.14]	0.14 [1.53]	0.17 [1.79]	0.23 [2.51]	0.25 [2.67]
Decile	NYSE	VW	0.36 [2.47]	0.12 [0.90]	0.29 [2.69]	0.29 [2.64]	0.36 [3.32]	0.35 [3.22]

Table 3: Conditional sort on size and EOB

This table presents results for conditional double sorts on size and EOB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EOB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EOB and short stocks with low EOB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EOB Quintiles					EOB Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.48 [2.18]	0.71 [3.10]	0.82 [3.27]	0.74 [2.97]	0.93 [3.41]	0.44 [3.70]	0.29 [2.60]	0.25 [2.23]	0.31 [2.73]	0.19 [1.64]	0.25 [2.16]
	(2)	0.63 [3.20]	0.80 [3.73]	0.83 [3.48]	0.78 [3.23]	0.84 [3.45]	0.21 [1.96]	0.06 [0.58]	0.12 [1.29]	0.10 [1.05]	0.09 [0.93]	0.07 [0.77]
	(3)	0.53 [3.12]	0.68 [3.38]	0.82 [3.85]	0.81 [3.61]	0.84 [3.62]	0.31 [2.56]	0.11 [0.98]	0.18 [1.94]	0.16 [1.72]	0.15 [1.59]	0.14 [1.44]
	(4)	0.57 [3.48]	0.61 [3.33]	0.69 [3.43]	0.73 [3.56]	0.80 [3.73]	0.23 [2.02]	0.04 [0.44]	0.16 [1.87]	0.15 [1.68]	0.17 [1.98]	0.16 [1.80]
	(5)	0.48 [3.35]	0.48 [3.01]	0.47 [2.81]	0.56 [3.08]	0.69 [3.62]	0.21 [1.61]	0.01 [0.07]	0.15 [1.42]	0.20 [1.87]	0.32 [3.02]	0.35 [3.25]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EOB Quintiles					EOB Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	418	420	419	418	417	32	33	33	36	40	
	(2)	116	116	116	115	116	59	59	59	60	59	
	(3)	83	82	82	82	82	102	101	101	101	101	
	(4)	68	68	68	68	68	216	214	212	212	206	
(5)	61	61	61	61	61	1564	1368	1405	1383	2078		

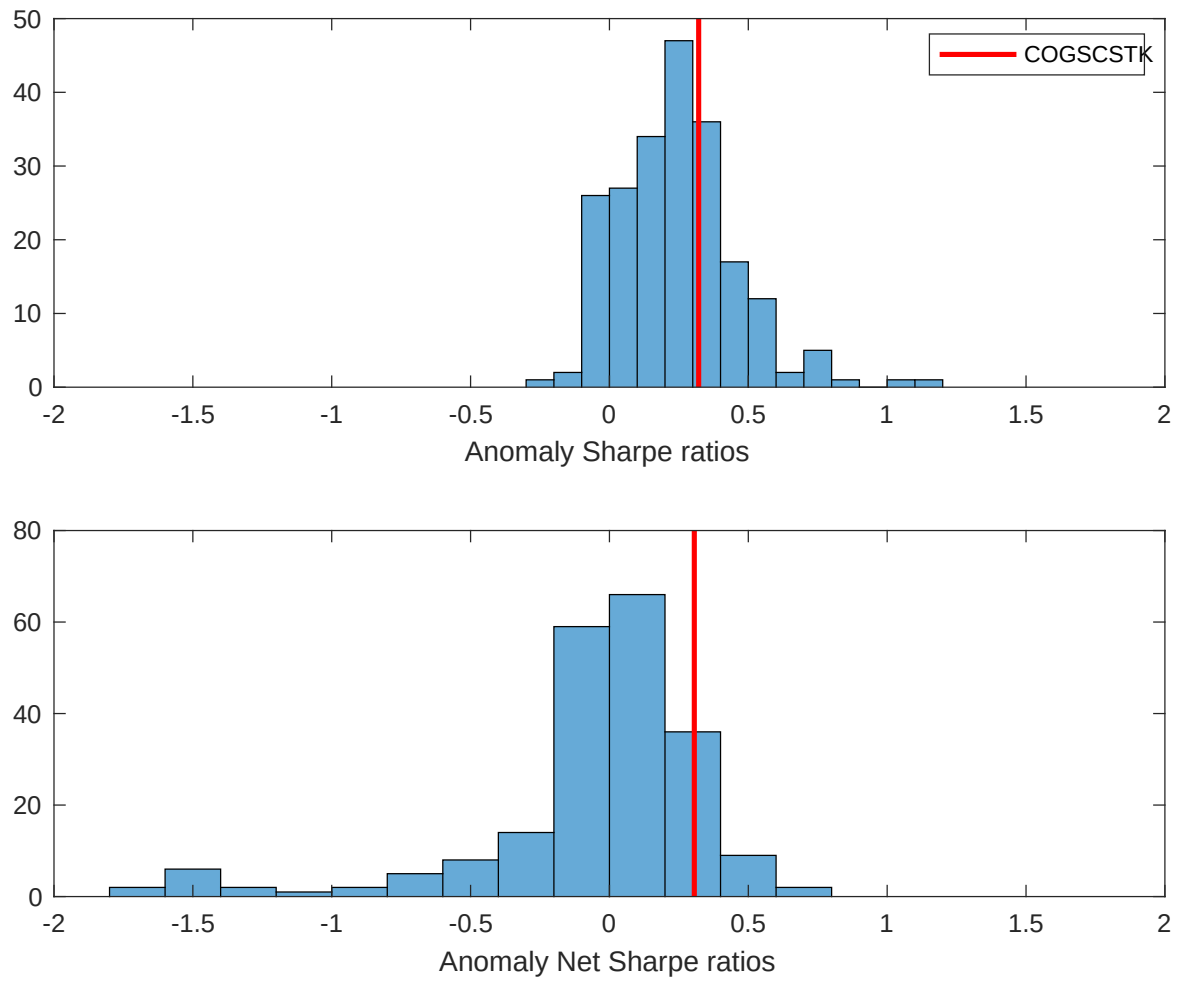


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EOB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

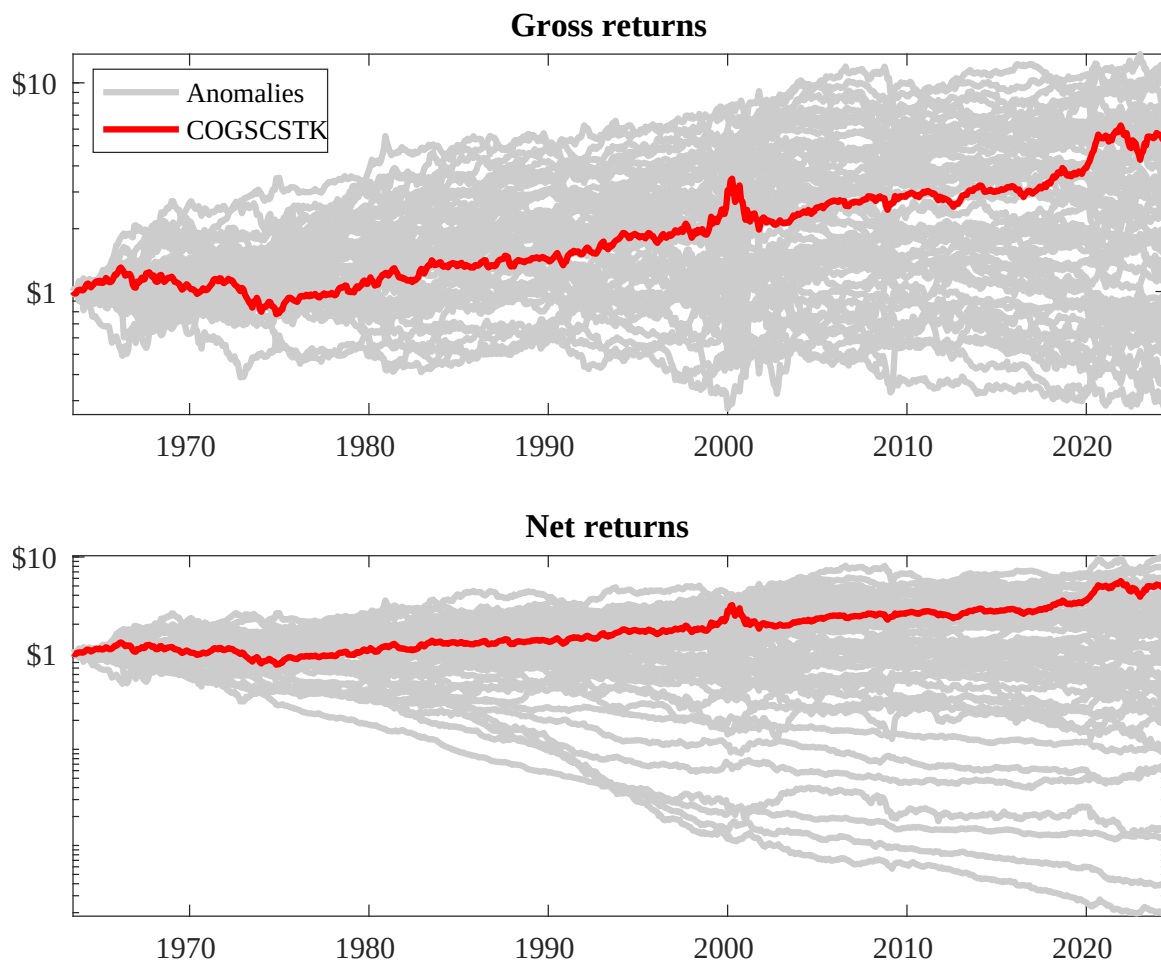
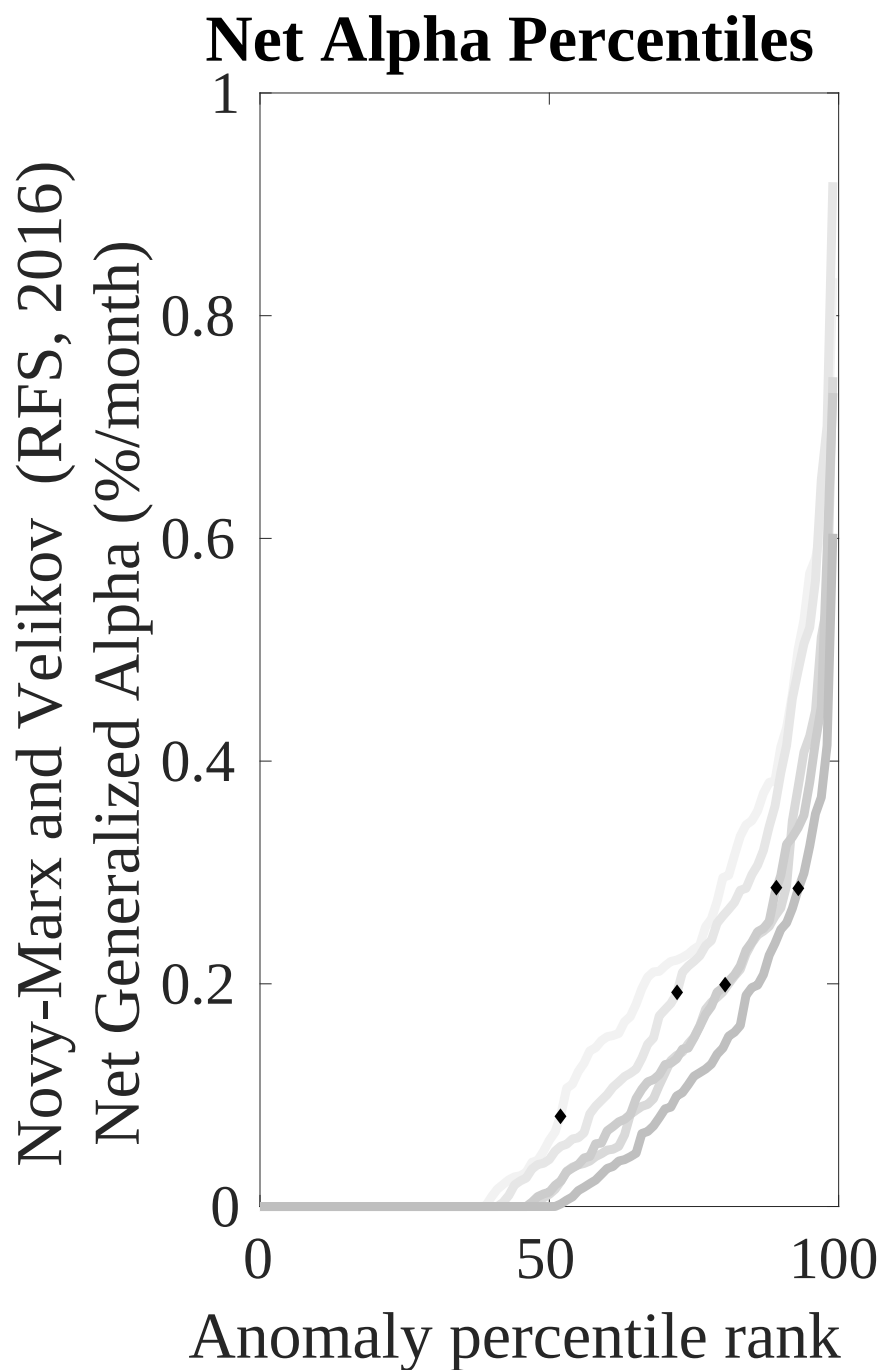
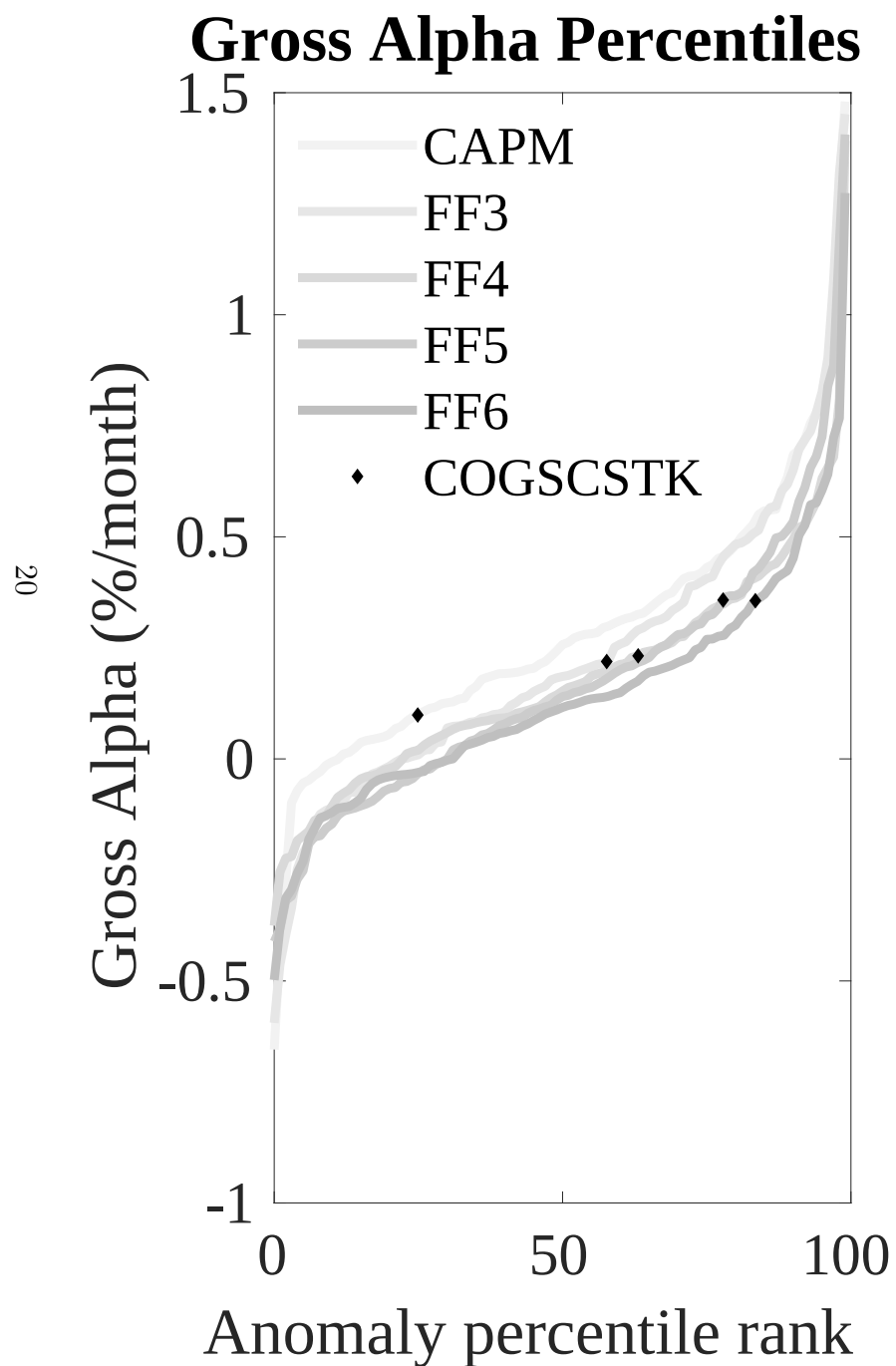


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EOB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



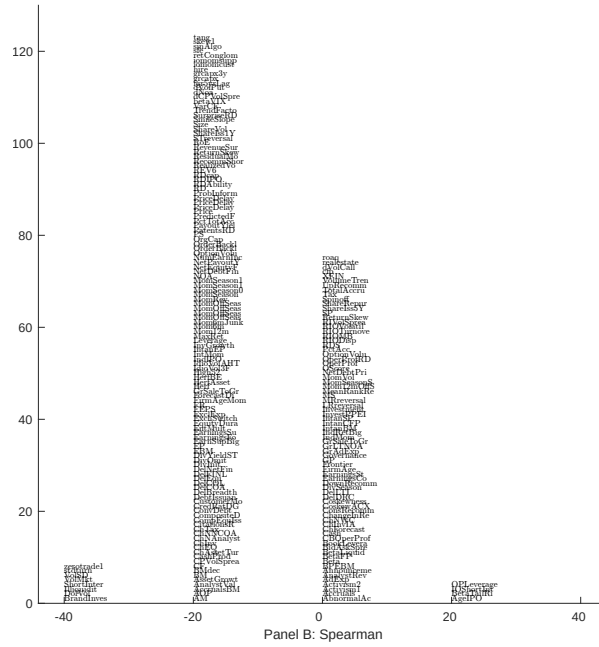
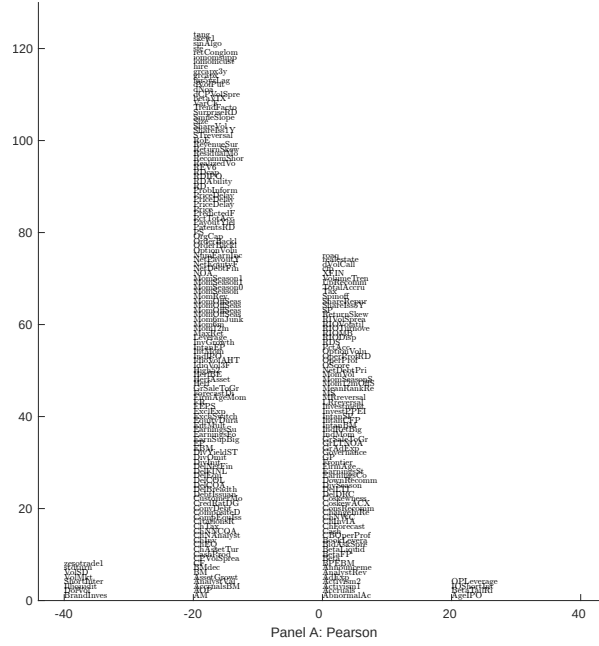


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EOB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

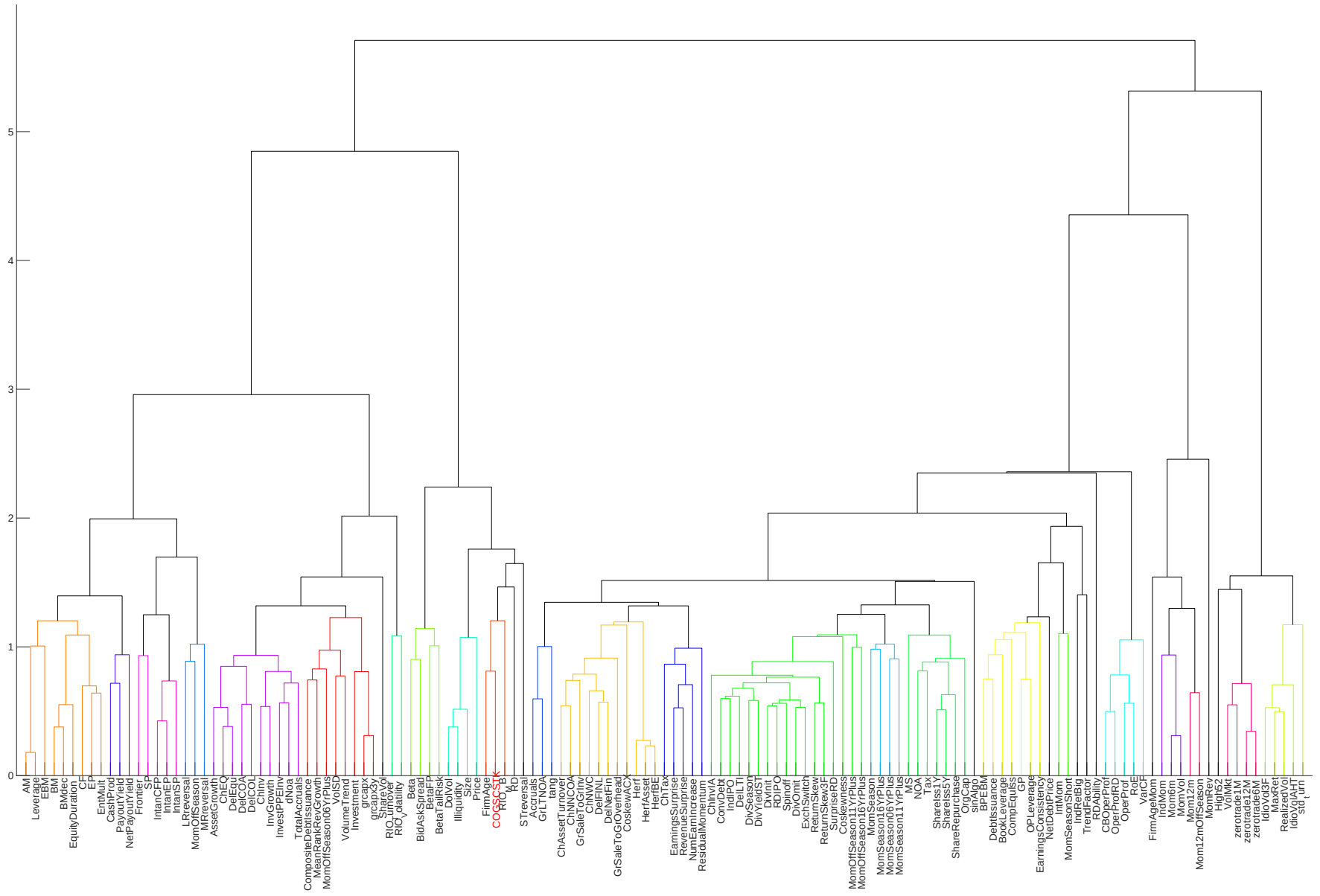


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

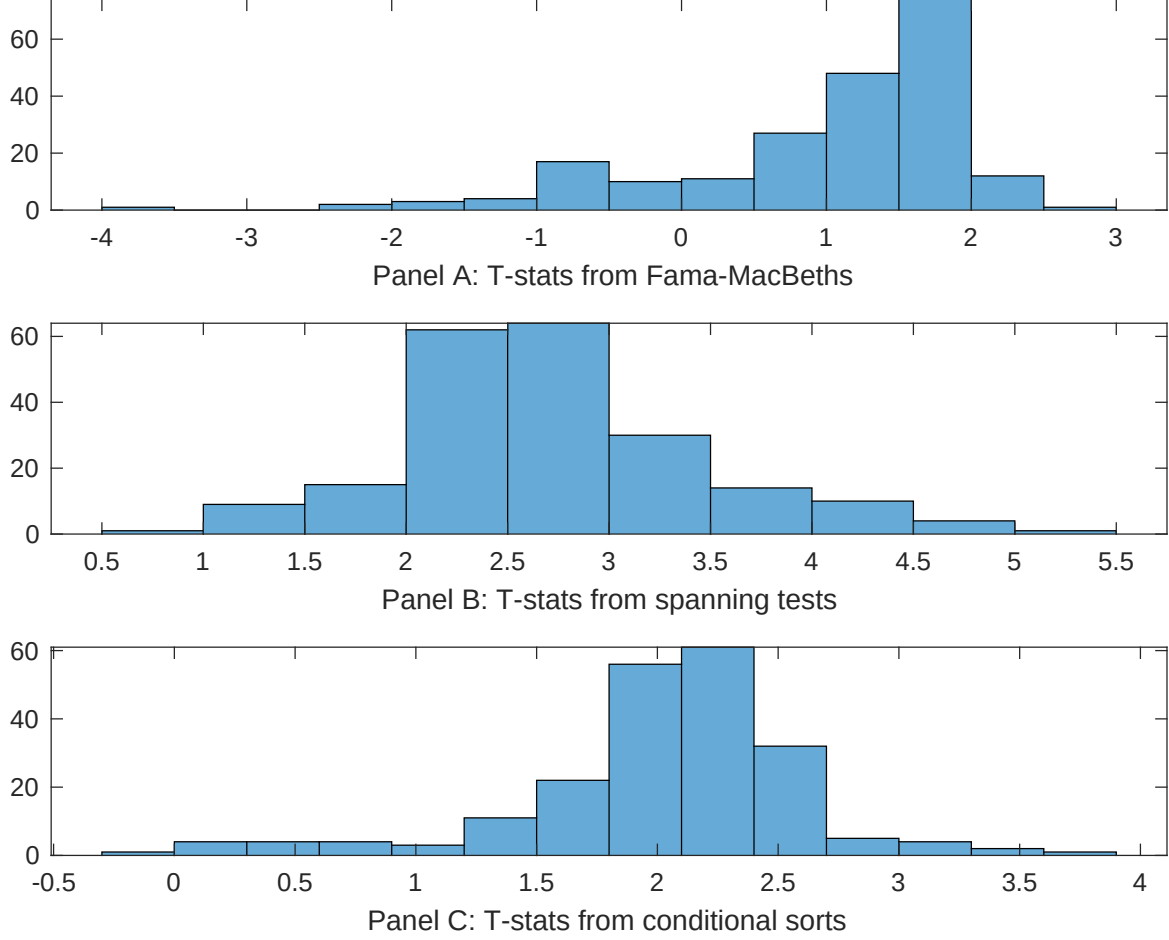


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EOB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EOB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EOB}EOB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EOB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EOB. Stocks are finally grouped into five EOB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EOB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EOB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EOB}EOB_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Tail risk beta, Predicted Analyst forecast error. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [7.36]	0.11 [4.76]	0.10 [4.61]	0.10 [4.64]	0.97 [6.05]	0.12 [5.33]	0.11 [6.28]
EOB	0.60 [1.40]	0.85 [1.74]	0.85 [1.76]	0.87 [1.79]	0.42 [1.17]	0.28 [1.12]	0.30 [1.27]
Anomaly 1	0.20 [1.53]						0.15 [2.47]
Anomaly 2		0.23 [1.38]					0.61 [0.26]
Anomaly 3			0.19 [2.87]				-0.65 [-0.72]
Anomaly 4				0.98 [2.98]			0.39 [0.23]
Anomaly 5					0.51 [3.13]		0.31 [1.70]
Anomaly 6						0.17 [0.65]	0.11 [0.58]
# months	727	727	727	727	727	492	487
$\bar{R}^2(\%)$	2	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EOB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EOB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Tail risk beta, Predicted Analyst forecast error. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.27 [3.16]	0.32 [3.70]	0.36 [4.15]	0.33 [3.83]	0.25 [2.88]	0.39 [3.87]	0.29 [3.03]
Anomaly 1	-27.60 [-10.54]						-13.40 [-1.97]
Anomaly 2		-23.99 [-9.25]					0.29 [0.04]
Anomaly 3			-22.17 [-8.46]				-1.67 [-0.17]
Anomaly 4				-23.15 [-8.80]			-9.70 [-1.15]
Anomaly 5					22.89 [9.12]		6.44 [1.98]
Anomaly 6						-24.06 [-7.94]	-15.02 [-4.81]
mkt	5.15 [2.20]	7.56 [3.26]	8.19 [3.49]	8.05 [3.46]	8.26 [3.61]	13.44 [5.49]	0.63 [0.23]
smb	8.22 [2.48]	15.80 [5.07]	21.14 [7.03]	23.73 [8.01]	15.50 [4.95]	23.63 [6.47]	11.71 [2.79]
hml	-18.82 [-4.80]	-20.35 [-5.12]	-20.53 [-5.12]	-21.00 [-5.27]	-28.66 [-7.25]	-14.95 [-3.15]	-14.86 [-3.26]
rmw	-6.79 [-1.63]	-6.68 [-1.56]	-7.31 [-1.68]	-7.97 [-1.86]	-13.13 [-3.22]	-9.87 [-2.07]	3.77 [0.78]
cma	-16.48 [-2.81]	-18.72 [-3.15]	-18.90 [-3.14]	-17.58 [-2.92]	-15.35 [-2.53]	-44.38 [-6.80]	-28.06 [-4.31]
umd	6.32 [3.07]	-0.05 [-0.03]	0.63 [0.31]	1.26 [0.62]	3.46 [1.70]	1.37 [0.60]	8.28 [3.14]
# months	728	728	728	728	728	492	488
$\bar{R}^2(\%)$	54	52	51	52	52	61	66

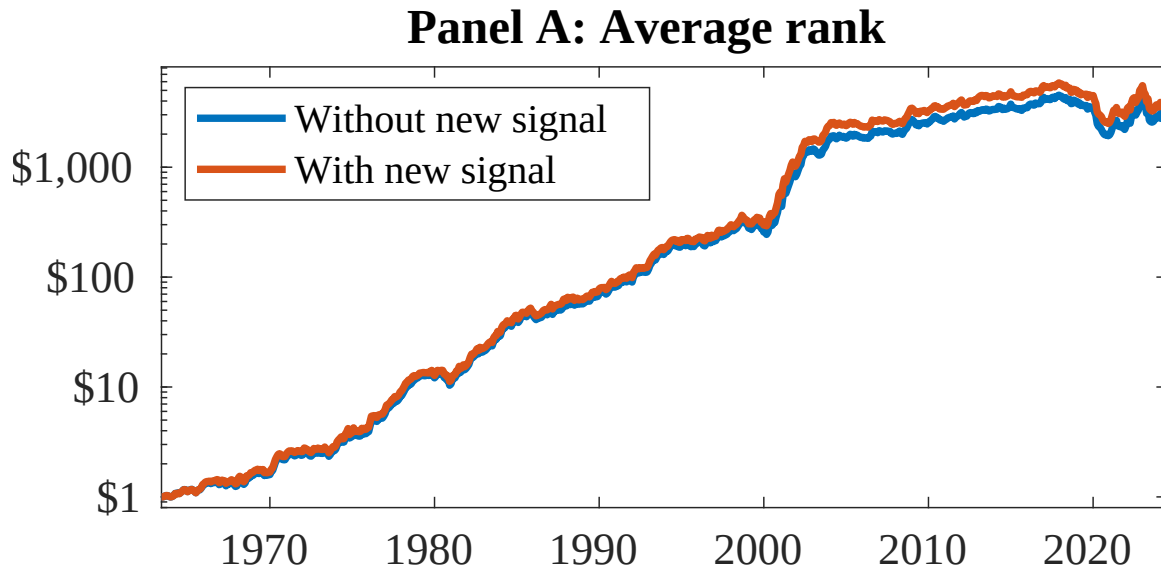


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EOB. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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